



Power-User Addendum

Developer Menu & Advanced Diagnostics Menu
Tester Engineering Suite v1.0.4-2026

This addendum documents the two hidden menus reserved for power users — the Developer menu (Ctrl+Shift+E) and the Advanced Diagnostics menu (Ctrl+Shift+A). Both menus are invisible by default. Both unlock with a keyboard chord and are intended for technicians who already understand what the items underneath them do.

Read the relevant section before clicking — some items are destructive.

Document not part of the standard user manual. Issued: 2026-05-01.

1. Developer Menu — Ctrl + Shift + E

Toggles a purple **Developer** item on the main menu strip. The menu is for inspecting and debugging the suite itself — license cache, transport state, telemetry, etc. Press the chord again to hide.

Session & State

Item	What it does
Show Current Session State...	Dumps VIN, TX/RX IDs, transport selection, J2534 firmware, frame counters, and DTC counters into a copyable report dialog.
Reset CAN-Known-Dead Flag	Clears the sticky 'CAN bus is unresponsive' decision so the next Live / Signal Analyser start re-probes from scratch.
Reset Current VIN	Wipes the in-memory VIN; next read overwrites it.
Show Active Transport	Reports user preference, currently active transport, K-Line state, and CAN-known-dead flag.

Connection

Item	What it does
List J2534 Devices	Full table of every J2534 PassThru DLL the device finder discovered, plus serial-port ELM327 entries.
Probe ECU Address...	Send an arbitrary service-ID byte to a chosen TX/RX pair and watch the response. Cheap reachability test before opening a full session.
Test Commands (PassThru / AT)...	Opens the raw J2534 / ELM327 console — manual PassThru calls, AT command stream, and the request/response log.
CAN Bus Health Check (J2284-3)...	Single-shot health probe — termination check, idle frame distribution, classifier per SAE J2284-3 §5.6 Table 1.
Check CAN Log for TES Fingerprint...	Scan a .log / .asc file for the HMAC fingerprint Tester Present writes during fingerprinted sessions. Used in forensics.
Fingerprinting active (toggle)	Master switch for CAN traffic fingerprinting. ON = 0x7E ISO-TP padding + 3127 ms TP cadence + 22 DE AD HMAC probe on connect. Per-vehicle escape hatch.
Idle CAN-Log Scanner (toggle)	Background filesystem watcher for fingerprint detection in third-party CAN logs. Refuses to start without EULA consent + IdleScanConsent registry flag = 1.
Idle Scanner Status...	Shows scanner state, last directory walked, file count, and any matches found this session.
Force-show Tesla tab	Forces the Tesla diagnostic tab visible without a Tesla VIN. For bench testing of Tesla diagnostics + flash UI.

License & Killswitch

Item	What it does
Show Cached RemoteConfig...	Reveals the DPAPI-decrypted contents of the offline license cache (server URL, API key, signing key, expiry).
Re-fetch RemoteConfig (online)	Forces a fresh online fetch from the license server and rewrites the local cache.

Clear License Cache	Wipes %LOCALAPPDATA%\TesterPresent\config.cache. Next launch starts cold and must reach the license server.
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Storage

Item	What it does
Open Vehicle History Folder	Explorer to ~/Documents/Tester Present History — birth certificates, flash certificates, receipts.
Open AppData / TesterPresent	Explorer to %LOCALAPPDATA%\TesterPresent — preferences.json, license cache, scratch files.
Open Build Output Folder	Explorer to bin/Debug or bin/Release — the directory containing the running EXE.

UI Diagnostics

Item	What it does
List Tab Pages	Every TabPage in the main TabControl with its index and visibility state. Useful for confirming VIN-gated tabs are showing/hiding correctly.
Test All Toast Types	Fires Info / Success / Warning / Error toasts in sequence so styling can be QA'd.
Force GC + Show Memory Stats	GC.Collect, then format working set + managed heap + GC generations. For chasing leaks during long sessions.
Force-show Tesla tab	(See Connection sub-menu).

Build Info

Item	What it does
Show Version & Build Info	Assembly version, build timestamp, target framework, runtime, OS, machine name.
Copy Diagnostic Bundle to Clipboard	Single text blob containing every diagnostic dump above — for pasting into a support ticket.

Power & Battery

Item	What it does
Show Power Status...	AC power state, battery percentage, lid state, and the active SetThreadExecutionState scope.
Open Windows Power Options	Shortcut to control.exe powercfg.cpl.
USB Suspend Field Notes...	Reference text on USB selective suspend behaviour for J2534 PassThru reliability.

2. Advanced Diagnostics — Ctrl + Shift + A

Toggles a dark-blue **Advanced Diagnostics** item on the main menu strip. Where the Developer menu inspects the suite, this menu inspects the *vehicle* — bus health, raw transport debugging, and structured fault-finding aids.

Bus & Transport

Item	What it does
CAN Bus Health Check (J2284-3)...	Single-shot diagnostic per SAE J2284-3 §5.6 Table 1. Reports termination, dominant-recessive ratio, error-counter status, and classifies each fault as Type A/B/C. Recommended first step when U-codes are present.
AT Commands / PassThru Console...	Raw transport console. Mode switches between J2534 PassThru calls (PassThruIoctl, PassThruWriteMsgs, etc.) and ELM327 AT commands (ATZ, ATRV, ATSP, ATMA). Use it to verify the device responds before suspecting the vehicle.
CANbus Diagnosis (live)...	Continuous frame stream — live counters for total frames, FPS, unique IDs, estimated bus load, and error counter. Top-20 dominant IDs table with hint annotations (OBD-II diag, 29-bit UDS, powertrain broadcast, body / comfort). Catches the slow-burn faults the single-shot J2284-3 test misses.

Fault Diagnosis

Item	What it does
Manual DTC Lookup...	Type or paste a code (P0420, U0100, B1601, C0035...) and get the J2012 family description, severity hint, and a triage walkthrough. Manufacturer-specific Ford codes are decoded against the bundled GMRDB; other OEMs return the generic family. Useful for preparing a quote before the car arrives.
Fault-Finding Walkthrough...	Wizard for the 'no idea where to start' cases. Pick a symptom from the left list (cranks-but-won't-start, intermittent stall, ABS light no DTCs, EVAP, battery drain, ...) and read the structured walkthrough — what to ask the customer, the cheapest first reading, the decision tree, and the trap most techs fall into.

3. Operating notes

Both menus are invisible by default. Show them with the keyboard chord, hide them with the same chord. No setting persists — re-launching the app hides them again. This keeps a regular operator from stumbling into them.

Build verification only. *The flash sequences in the Tesla tab were reverse-engineered and have not been validated against real hardware. The Advanced Diagnostics menu's read-only tools (bus health, manual DTC lookup, fault-finding walkthrough) are safe. The PassThru / AT console can issue arbitrary CAN traffic — treat it like a torque wrench, not a hammer.*

Where to find more. Detailed protocol notes are in the project AGENTS.md. DTC databases used by Manual DTC Lookup are bundled (Ford GMRDB) and decrypted at runtime via the license-server-issued AES keys; without an active license the lookup falls back to generic SAE J2012 family decoding.